

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

INTRO TO CRITICAL INFRASTRUCTURE FOR IT LEARNERS

LAB 07 — EXPLORING ICS PACKET CAPTURES

Student Edition

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Lab 07: Exploring ICS Packet Captures

Executive Summary

Analysis of the provided ICS packet capture identified multiple industrial and IT protocols including HTTP, FTP, Modbus/TCP, EtherNet/IP, and Siemens S7. Several security weaknesses were observed that could expose industrial assets, credentials, and operational information to unauthorized users.

Key Findings

HTTP Basic Authentication Analysis

After applying the **http.authbasic** filter in Wireshark, several HTTP requests using Basic Authentication were identified. The traffic showed communication between **192.168.2.111** and **192.168.88.115**. By examining the HTTP Authorization header, it was observed that authentication details were transmitted using Base64 encoding.

Wireshark was able to decode the authentication information, revealing that user credentials were being sent across the network without encryption. This demonstrates a significant security concern because Base64 encoding can be easily reversed, allowing anyone who captures the traffic to recover usernames and passwords.

Security Observation:

The use of HTTP Basic Authentication without encryption exposes credentials to interception and should be replaced with secure authentication mechanisms protected by HTTPS.

HTTP POST Request Analysis

The filter **http.request.method == "POST"** was applied to examine HTTP form submissions within the capture. Multiple POST requests were identified from **192.168.2.111** to **192.168.88.115**, including requests to pages such as `/status`, `/goform/svLogin`, `/home.asp`, and `/view/`.

Inspection of the packet contents revealed that login information was being transmitted within the form data. Usernames and passwords were visible in the requests, indicating that sensitive information was being sent without adequate protection.

Security Observation:

Exposed login credentials in HTTP POST traffic can be intercepted by attackers with access to network communications. This highlights the importance of encrypting web traffic using HTTPS.

FTP Traffic Analysis

Using the **ftp** display filter, an FTP session was observed between **10.10.10.50** and **192.168.2.206**. The traffic contained authentication commands and server responses that were visible in plain text.

The capture revealed a login attempt using the username **cameraadmin**. The FTP server banner identified the device as an **AXIS 206 Network Camera running FTP Server version 4.40**. Additional information, such as the operating system type and directory structure, was also visible within the session.

Security Observation:

FTP does not provide encryption for usernames, passwords, or transferred data. In addition, device information exposed through server banners can assist attackers in identifying potential vulnerabilities.

Modbus Traffic Analysis

The **modbus** filter was used to examine industrial communications between **192.168.2.111** and **10.10.20.12**. Several Modbus/TCP operations were identified on TCP port **502**.

The captured traffic included multiple function codes, such as:

- Function Code 1: Read Coils
- Function Code 3: Read Holding Registers
- Function Code 6: Write Single Register

- Function Code 43: Read Device Identification

These operations demonstrate both monitoring and control activities commonly performed within industrial automation systems.

Security Observation:

Modbus provides no native authentication or encryption. As a result, communication can be monitored, modified, or spoofed by unauthorized parties if proper network protections are not in place.

DNP3 Traffic Analysis

A search was conducted using the **dnp3** filter to identify Distributed Network Protocol (DNP3) communications within the packet capture.

If No DNP3 Packets Were Found:

No DNP3 traffic was identified in the provided packet capture. Therefore, no DNP3 communication session could be analyzed during this exercise.

DNP3 is widely used in SCADA and utility environments, particularly in electric power, water treatment, and other critical infrastructure sectors. The protocol enables communication between master stations and remote devices for monitoring and control purposes.

Security Observation:

Although no DNP3 traffic was present in this capture, traditional DNP3 implementations are often deployed without encryption, making network segmentation and monitoring important security controls. [*Find the relevant evidence in Snapshots folder*](#)

Complete Protocol Evidence Matrix

Protocol	Filter	Source	Destination	Observed Operation	Security Significance
HTTP Basic Auth	http.authbasic	192.168.2.111	192.168.88.115	GET /view/	Credentials recoverable from Base64 encoding
HTTP POST	http.request.method == "POST"	192.168.2.111	192.168.88.115	Login form submission	Username and password exposed
FTP	ftp	10.10.10.50	192.168.2.206	USER/PASS authentication	Cleartext credentials and device fingerprinting
Modbus/TCP	modbus	192.168.2.111	10.10.20.12	Read Coils, Read Holding Registers, Write Register	No authentication or encryption
EtherNet/IP	enip	10.10.10.50	10.10.20.11	Register Session	Industrial session information exposed
S7	s7comm	10.10.10.50	10.10.20.11	Read Var (DB0.DBX0.0)	PLC data visible in network traffic

Knowledge Check

1. Why does Base64 encoding not provide confidentiality for HTTP Basic authentication?

Response: Base64 is an encoding method, not an encryption mechanism. Anyone who intercepts the HTTP traffic can easily decode the Base64-encoded username and password using common tools. Therefore, HTTP Basic Authentication does not provide confidentiality unless it is protected by HTTPS/TLS.

2. What information can device banners or protocol metadata reveal to an analyst?

Response: Device banners and protocol metadata can reveal information such as device type, vendor name, firmware version, operating system, network addresses, protocol versions, and system roles. This information helps analysts identify assets but can also assist attackers in targeting known vulnerabilities.

3. How do Modbus, DNP3, EtherNet/IP and S7 differ from ordinary user-facing protocols such as HTTP or FTP?

Response: Modbus, DNP3, EtherNet/IP, and S7 are industrial communication protocols designed for communication between control systems, PLCs, RTUs, sensors, and SCADA systems. Unlike HTTP and FTP, which are used for web and file-transfer services, industrial protocols focus on monitoring and controlling physical processes in critical infrastructure and industrial environments.

4. Why is passive packet analysis valuable for understanding an ICS environment?

Response: Passive packet analysis allows security analysts to observe network communications without transmitting traffic or interacting with devices. This reduces the risk of disrupting industrial operations while providing visibility into network architecture, device communications, protocols, and potential security issues.

5. What compensating controls can reduce risk where legacy industrial protocols cannot be replaced immediately?

Response: Organizations can reduce risk by implementing network segmentation, firewalls, industrial DMZs, access control lists (ACLs), VPNs, intrusion detection systems (IDS), continuous monitoring, strong authentication, and strict access management. These controls help protect legacy protocols that lack built-in security features.